Counting & Sampling

- If you have discrete prob. space with all outcomes in Ω being equally likely, then for any event A

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}$$

Basic rules

→ If we have n choices $c_1, c_2 \ldots c_n$ & each c_i has m_i options then # possibilities for tuple $(c_1, c_2 \ldots c_n)$

: $m_1 \times m_2 \ldots \times m_n$

Eg 4 shirts, 3 pants no of ways of dressing $= 4 \times 3 = 12$

Eg Coin toss 10 times # Config $= 2^{10}$

→ n integers, all distinct.

# of orderings $= n \times (n-1) \ldots 1 = n!$

→ Let S, # subset of $S = 2^{|S|}$ (each element in subset or not)

Sampling Models

Urn / bin has m distinguishable balls. Pick n balls from urn

Various ways of drawing / sampling:

① With replacement & ordering ({red, blue} not same as {blue, red})

O/p is an ordered tuple $(a_1, a_2 \ldots a_n)$. Each a_i has m possibilities. # configs possible : m^n

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(2) Without replacement & with ordering.

Now get a tuple $(a_1, a_2 \ldots a_n)$ where $a_i \neq a_j$ [Need $m \geq n$ here]

possibilities $= m(m-1) \ldots (m-n+1) = \frac{!m}{!(m-n)}$

(3) <u>Without replacement & without ordering</u>

Now, o/p is a set $\{a_1, a_2 \ldots a_n\}$ [don't care about order]

Say X is # of possibilities. For each set of this type, there are $n!$ possible orders.

Thus from (2), $X \times !n = \frac{!m}{!(m-n)} = \binom{m}{n}$

(4) <u>With replacement & without ordering</u>

Now, o/p is the tuple $(n_1, n_2 \ldots n_m)$ $(n_i \geq 0)$ where n_k = # of balls of type k drawn & $\sum_{i=1}^{m} n_i = n$.

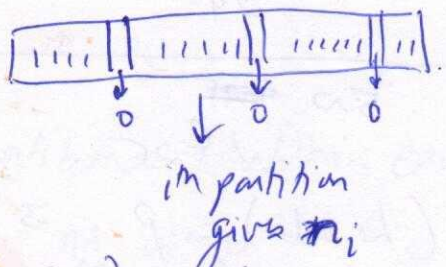
So we need cardinality of set

$$Z = \{ \bar{k} \in (\mathbb{N} \cup \{0\})^m, \sum k_i = n \}$$

$$|Z| = {}^{n+m-1}C_{m-1}$$

Pf

Assume you have a bit string of length $n+m-1$, with $m-1$ zeros & n ones



Need to select $m-1$ positions from $n+m-1$ positions.

So get $|Z| = {}^{n+m-1}C_{m-1}$

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Eg 52 cards randomly shuffled. Prob that 4 aces are in a row.

If 4 aces clubbed together into a card, 49 cards now.

$$\mathbb{P} = \frac{49! \times 4!}{52!} \qquad |\Omega| = 52! \quad |A| = 49! \times 4!$$

Eg - Fair die, Roll 6 times

P(all rolls diff)

$$|\Omega| = 6^6 \qquad |A| = 6!$$

$$\mathbb{P} = \frac{6!}{6^6}$$

Eg Seller has 500 comp. for sale. 15% faulty. Buyer picks 25 at random and buys only if atmost 2 are defective. $\mathbb{P}$(seller makes sale) = ?

No of defective $= \frac{15}{100} \times 500 = 75$

Sell made = 0 def selected + 1 def select + 2 def select

$$= \frac{{}^{425}C_{25} + {}^{425}C_{24} \times {}^{75}C_1 + {}^{425}C_{23} \times {}^{75}C_2}{{}^{500}C_{25}}$$

Eg Biased coin $\mathbb{P}$(heads) = p. 3 ind. tosses

$\mathbb{P}$(2 heads)

$\mathbb{P}$(any given seq of k heads, n-k tails) $= p^k (1-p)^{n-k}$

$\mathbb{P}$(~~2 heads 1 tails~~ k heads) $= \binom{n}{k} p^k (1-p)^{n-k}$ (Binomial Prob)

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Eg Coin toss - $P(\text{heads}) = p$

Given 3 heads in 10 tosses, what is prob. that first 2 tosses were heads?

$A \to$ first 2 tosses Heads, $B \to$ 3 heads in 10 tosses

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{{}^8C_1}{{}^{10}C_3}$$

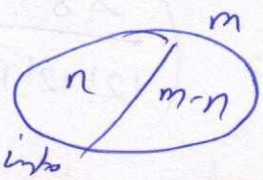
$$\mathbb{P}(A \cap B) = p^2 \cdot {}^8C_1 \cdot p(1-p)^7$$

$$\mathbb{P}(B) = {}^{10}C_3\, p^3 (1-p)^7$$

<u>Partitions</u> - Suppose you choose n objects out of m ~~distinct~~ distinct obj.

No of ways $= \binom{m}{n} = \frac{!m}{!n\,!m-n}$

Same as partitioning the set of m objects into two partitions: selected & unselected



<u>Generalization</u> - Suppose want to divide into r parts s.t i^{th} partition gets n_i elements

Divide n items into $n_1, n_2 \ldots n_r$ s.t $\sum_{i=1}^{r} n_i = n$

Suppose X such partitions. Within each partition i, can arrange items in $n_i!$ ways

$$X \times (n_1!) \times (n_2!) \ldots (n_r!) = n!$$

Every overall permutation can be mapped to a combn

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Thus $X = \underbrace{\dfrac{n!}{n_1!\, n_2! \cdots n_r!}}_{\text{Multinomial coefficient}}$

Eg Deck of 52 cards, dealt fairly to 4 players.

$\mathbb{P}$(each player gets an ace).

~~Total no of shuffling = 52!~~ [~~First player takes 1st 13, 2nd player next 13 ...~~]

~~Total no of ways of dealing~~

ways of dealing 13 cards each to 4 player

$$= \frac{52!}{13!\,13!\,13!\,13!} \quad (A)$$

ways of dealing such that one ace to each

$$= \left(\frac{48!}{12!\,12!\,12!\,12!}\right) \times \frac{4!}{1!\,1!\,1!\,1!} \quad (B)$$

$$\mathbb{P} = \frac{B}{A}$$

Eg – n items, k def.. Random permutation

$\mathbb{P}$(First def. encountered at position 3)

$$= \frac{\binom{n-k}{3-1}\, 3-1 \cdot \binom{k}{1} \cdot (n-3)!}{n!}$$

$$\frac{(3-1) \cdot (\quad)}{\downarrow\ n-3}$$

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